Sahaj Bhadja

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Summary

Performance-focused software developer with strong **C++**, algorithms, and systems fundamentals. Experienced optimizing concurrent pipelines, reducing latency, and building correct low-level software. Eager to learn proprietary trading systems and contribute to a **low-latency algorithmic trading platform** in a fast-paced environment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Operating Systems, Data Structures, Object-Oriented Programming, Computer Networks

Experience

USC Dornsife **Los Angeles, CA**
Software Developer *September 2025 – Present*
- Optimized production services supporting 500+ users, cutting dashboard latency from **1.2s to 350ms** through indexing, query tuning, and performance-conscious access patterns across concurrent background workloads in reliability-sensitive environments.
- Restructured fault-tolerant asynchronous job execution with explicit state transitions and failure isolation, improving throughput by **3x** under concurrent load while maintaining **99%+ reliability** across **5K+ monthly** integration jobs.

Grad8 **Los Angeles, CA**
Software Developer Intern *June 2025 – August 2025*
- Built concurrent batch-processing pipelines using **multithreading** and **ThreadPoolExecutor**, handling **10K+ daily requests** with fault-tolerant execution, stable peak-load performance, and detail-oriented debugging across production API workflows.
- Collaborated with senior engineers on performance tuning and release readiness, reducing feedback turnaround by **60%** while learning quickly in a fast-paced startup environment with rigorous code review standards.

Pandit Deendayal Energy University **India**
Software Developer - Research Fellowship *January 2023 – May 2024*
- Developed and benchmarked novel **consensus algorithms** in simulation environments, reducing centralized-control and risk exposure by **25%** through careful algorithm design, correctness analysis, and performance-oriented implementation tradeoffs.
- Contributed to peer-reviewed systems research presented at IISc Bengaluru, strengthening ability to learn complex methodologies, communicate technical results, and pursue new opportunities in algorithm-heavy software domains.

Aasma Technology Solutions **India**
Software Developer - (SecureSync) *August 2023 – July 2024*
- Engineered performant API modules using **OOP design patterns** (Singleton, Factory), improving response time by **30%** through algorithmic optimization, caching, pagination, and low-bandwidth payload reduction across distributed microservice integrations.
- Maintained **20+ automated test suites** across Windows and **Linux** environments in CI pipelines, increasing release confidence by **68%** while collaborating with senior developers in Agile sprints on detail-oriented production delivery.

Aasma Technology Solutions **India**
Software Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built real-time synchronization services with **WebSockets** across **2M+ records**, delivering sub-second updates with **33% fewer conflicts** through careful concurrency handling, structured data modeling, and performance tuning on operational dashboards.
- Improved API performance via caching and database latency reduction, automating Docker/Jenkins release flows that cut deployment cycles by **50%** while working alongside experienced developers in a fast-paced delivery environment.

Skills

C++ & Systems: **C++**, modern C++ (smart pointers, STL), multi-threading, concurrency, OOP, design patterns, memory management, performance optimization, low-latency systems, Linux, Bash, debugging, profiling (familiar)
Algorithms & Trading: data structures, algorithms, analysis of algorithms, operating systems, real-time systems, distributed systems, consensus algorithms, algorithmic trading interest, financial systems (familiar), market data (familiar)
Engineering: Python, Java, SQL, Git, TCP/IP, sockets (familiar), fault tolerance, code reviews, Agile, fast-paced learning, detail-oriented delivery, cross-functional collaboration, production reliability

Projects

CPU Scheduling Simulator Live | *C++, Algorithms, Performance Engineering*
- Implemented preemptive scheduling algorithms (FCFS, SJF, SRT, RR, MLQ) in **C++** with correctness testing across **1K+** process combinations, reducing average turnaround time by **25%** and debugging time by **20%** through performance engineering.

FinNexus Live | *Real-Time Financial Systems, WebSockets, High Throughput*
- Built a fintech platform handling **1M+ transactions** for **50K+ users** with real-time WebSocket updates, secure tokenized access, and scalable architecture supporting low-latency financial data visibility across high-volume operational workflows.

ExecPath Profiler | *C++, Concurrency, Low-Latency Simulation*
- Developed a C++ simulation harness for concurrent event processing with timing metrics, queue-depth analysis, and bottleneck identification, applying systems-level thinking to optimize execution paths and validate correctness under bursty high-throughput workloads.